Evolutionary Dual-Ensemble Class Imbalance Learning for Human Activity Recognition

Yinan Guo, Yaoqi Chu, Botao Jiao, Jian Cheng, Zekuan Yu, Ning Cui, and Lianbo Ma

***Abstract*—Human activity recognition is an imbalance classification problem in essence since various human actions may occur at different frequencies. Traditional ensemble class imbalance learning methods integrate resampling technique with multi-classifier models, which obtain better generalization than single ones. However, the number of base classifiers is often determined in advance, resulting in the redundant structure of a multi-classifier and larger computational cost. Moreover, various combinations of base classifiers may have similar recognition accuracy, forming a multimodel optimization problem. To address the issue, a dual-ensemble class imbalance learning method is presented, in which two ensemble learning models are nested each other. For each sub-dataset, three heterogeneous sub-classifiers compose of internal ensemble learning model, and the one with the highest recognition accuracy is selected as base classifier. In external ensemble learning model, an optimal combination that contains the smallest number of base classifiers and identifies human actions most exactly is found by multimodal evolutionary algorithms. Based on seven imbalanced activity datasets derived from localization data for person activity, statistical experimental results show that the proposed evolutionary dual-ensemble class imbalance learning method provides the simplest ensemble structure with the best accuracy for imbalanced data that outperforms other five widely-used ensemble classification methods on all metrics.**

***Index Terms*—Class imbalance learning, ensemble learning, under-sampling, multi-modal evolutionary algorithm, human activity recognition.**

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I. Introduction

Human activity recognition (HAR) is an important and hot topic in the field of artificial intelligence due to its wide applications in smart home, intelligent medical treatment and sports [1], [2]. In general, a person's actions can be identified from images or videos collected from cameras, and data obtained from various sensors. Correspondingly, the methods for identifying human behaviors can be divided into two categories, including vision- and sensor-based recognitions. The former distinguishes various actions by traditional image process methods or deep learning, which is intuitive, but easily affected by light conditions, resolution and the other noises. Compared with video-based data, collecting data from mobile or wearable sensors is convenient, fast and cheap. Thus, it becomes universally available for many scenarios. For example, the number of steps taken in walking is counted in terms of recorded movement tracks [3], and wrestling is employed to judge whether a person has fallen down [4]. The commonly-used sensors for monitoring human motions, such as accelerometer, inertial navigation, and gyroscope, can be fixed on different parts of the body, including waist, wrist, chest, thigh and so on.

Based on the motion data retrieved from various monitoring resources, traditional machine learning methods are introduced to identify human behaviors [5]. Li, *et al.* [6] build a hidden Markov model to classify the data gathered from accelerometer. Qian [7] mapped acceleration data into three-dimensional feature space and recognized the motion by a voting method. Su [8] intercepted a monitoring data via the last-bit matching. The training samples are segmented, and 2-norm accumulation of global gradient vector is employed as the matching metric. Banos [9] further discussed the influence of sliding-window size on recognition accuracy of behaviors, and presented a selection criteria for its optimal size. Though decision tree-, support vector machine-, naive Bayes-, and hidden Markov-based classifiers [10]–[13] have achieved the satisfied recognition rate, there exists some difficulties in the field of human behavior recognition. One is that training classifiers depend on the features that manually selected in time- or frequency-domain, which is time-consuming and not objective enough. In particular, inappropriate features may result in misclassification for complex behaviors. The other is that the number of data corresponding to various behaviors may be skewed. For example, an adult seldomly falls down, but lies down many times. Hence, the falling and lying down (FLD) dataset derived from localization data for person activity (LDPA) datasets that contains both of above-mentioned human actions is imbalanced. The motion with the larger number of data is called the majority class, whereas other occurred infrequently belongs to the minority class that is easily misclassified by traditional machine learning methods, resulting in lower classification accuracy.

Compared with traditional machine learning, deep learning automatically extracts features instead of manual selection of traditional methods during the training, which describes the characteristics of human activities in complex scenes more accurately. Nakano [14] captured the dynamic features of time series by recursive graph, and classified them by a convolutional neural network (CNN) so as to identify dynamic activities, such as walking, going upstairs and going downstairs. Yin [15] also employed CNN to extract local features, and analyzed the influence of time series's length on recognition accuracy. How, *et al.* [16] trained a recurrent neural network (RNN) by human activity data, with the purpose of recognizing the current action. However, it obtains worse accuracy for longer time series due to gradient disappearance. To solve the problem, Yu [17] established long short-term memory (LSTM) network, and automatically selected the correlation features that have a significant effect on classification. Pienaar [18] integrated two above methods to form a LSTM-RNN-based recognition model, which obtained good generalization for predicting six kinds of behaviors in time. However, deep learning methods generally depend on a large amount of human activity data. Thus, it has high computational complexity and limited performance on imbalanced scenes.

Without loss of generality, minority samples are more important and also carry a higher misclassification cost [19]. To address the issue, rich studies have done on class imbalance learning, which can be classified into three categories, including data-, algorithm- and hybrid level methods. Data-level classification methods convert imbalanced datasets to balanced ones by over-sampling minority samples or under-sampling majority ones [20], [21]. Synthetic minority over-sampling technique (SMOTE), as a typical one, generates minority samples nearby the decision boundary to improve classification accuracy. At the algorithmic level, some special techniques, such as adjusting the misclassification cost of each class, or changing the classification threshold, are designed in imbalance classification algorithms [22].

Being different from them, ensemble imbalance learning integrates resampling techniques with multi-classifier model, which is grounded on the assumption that synthesizing the outputs of multiple weak classifiers can obtain better generalization than a single one. In order to mine the implicit distribution of data and build a more rational classifier for each sub-dataset, bagging and boosting, as the representative ensemble class imbalance learning, integrate multiple base classifiers to reduce the overall classification errors [23]–[25]. Each base classifier is trained by a balanced sub-dataset randomly extracted from the original skewed one after resampling. Though various sub-datasets are employed for independently training, the similar, even same base classifiers may be produced. In addition, the number of base classifiers is often determined by human in advance, resulting in the redundant structure of a multi-classifier, which increases the computational cost and cannot meet the real-time requirements of human behavior recognition [26]. Thus, determining the rational number of base classifiers, building a better learning model for each sub-dataset and maintaining the good diversity of a multi-classifier are three key issues of ensemble imbalance classification.

To address these issues, an evolutionary dual-ensemble class imbalance learning method (ED-Ensemble) is proposed, in which base classifiers employ various learning methods and multimodal evolutionary algorithm is introduced to find the most compact structure with the highest recognition accuracy for an ensemble classifier of human activity recognition. This paper has the following three-fold contributions:

1) A dual-ensemble framework composed of internal and external ensemble learning models is constructed. Internal ensemble learning model is designed to seek an optimal base classifier for each sub-dataset obtained by a resampling strategy, with the purpose of reducing computational cost while maintaining the highest recognition accuracy. In external ensemble learning model, the optimal combination with the simplest ensemble structure is developed based on base classifiers.
2) In internal ensemble learning model, three kinds of learning methods, including extreme learning machine (ELM), Naive Bayesian model (NBM) and K-nearest neighbor (KNN), are employed to build sub-classifiers, with the purpose of enlarging the difference among them. For each sub-dataset, the sub-classifier with the best recognition accuracy is selected as a base classifier.
3) In external ensemble model, multimodal evolutionary algorithm based on niche strategy is introduced to seek the optimal combinations of base classifiers in terms of the classification performance. G-means of K-fold cross-validation is employed as the objective function. Among the potential optimum, the combination aggregating the smallest amount of base classifiers by weighted sum provides the most compact structure for external ensemble learning model.

The rest of the paper is organized as follows. The framework of Evolutionary Dual-ensemble class imbalance learning is illustrated in Section II. In Section III, hybrid base classifier of internal ensemble learning model is constructed. Section IV introduces multimodal evolutionary algorithm to seek the most compact structure of external ensemble learning model. Experimental results are compared and analyzed for a real-world imbalanced human activity dataset in Section V. Finally, Section VI concludes the whole paper, and plans the topic to be researched in the future.

II. The Framework of Evolutionary Dual-Ensemble Class Imbalance Learning

Evolutionary Dual-ensemble class imbalance learning employs a nested structure to seek the smallest number of hybrid base classifiers that achieve the highest recognition rate, as shown in Fig. 1, with the purpose of decreasing computational cost while keeping the highest classification accuracy.

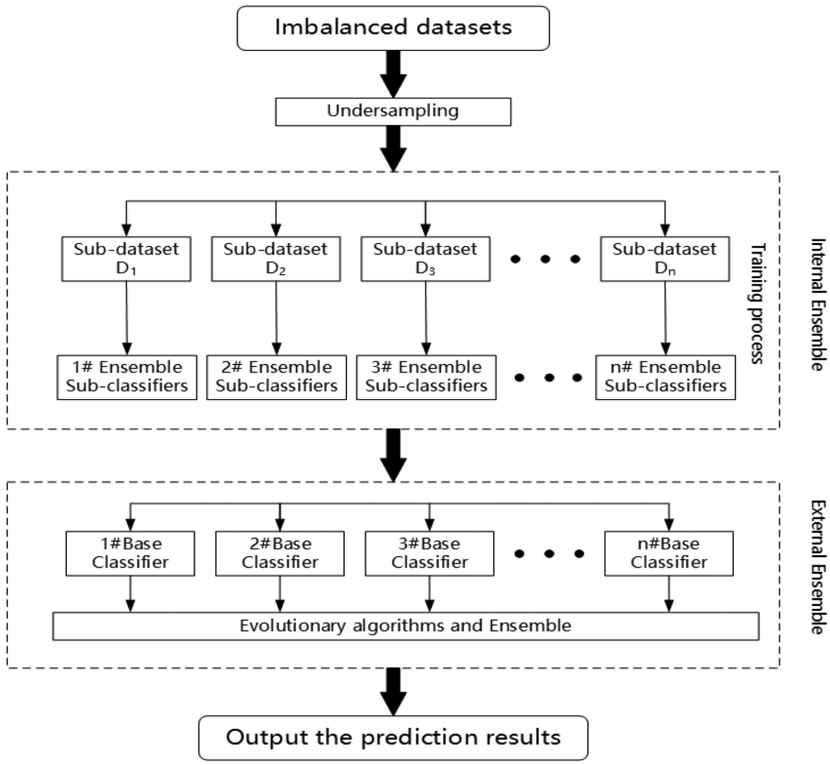


Fig. 1. The framework of Evolutionary Dual-ensemble class imbalance learning.

In internal ensemble learning model, each sub-dataset is employed to train more than one heterogeneous sub-classifiers. Among multiple sub-classifiers, the one with the optimal recognition performance is selected as a base classifier. Following that, an optimal portfolio model of base classifiers is found in external ensemble learning in terms of recognition accuracy and portfolio size.

In internal ensemble learning model, an appropriate base classifier is generated for each sub-dataset after under-sampling. Without loss of generality, commonly-used ensemble imbalance classification methods consist of homogeneous sub-classifiers that only capture the local characteristics of an imbalanced dataset, which limits the overall classification accuracy [27]. To tackle the drawback, more than one heterogeneous sub-classifier is employed to form an ensemble learning model for a sub-dataset, and then an optimal one with the highest recognition accuracy is selected as a base classifier. In ED-ensemble method, ELM, NBM and KNN are introduced to construct sub-classifiers due to their strong explanation, with the purpose of sketching the high-order information of a sub-dataset for human activities, and increasing the diversity among them.

To the best of our knowledge, the number of base classifiers are generally set in advance before training. However, too few base classifiers will deteriorates their diversity, resulting in worse classification accuracy because each base classifier only sketches the local feature of an imbalanced dataset. On the contrary, the computation cost of a multi-classifier with larger scale is high and more worthless base classifiers may be built [28]–[30]. Assume that n_{I} is the number of optional base classifiers produced by internal ensemble learning model and n_{O} is that of ones selected for building the final ensemble learning model. An optimal combination is obtained after seeking $\sum C_{n_{\mathrm{I}}}^{j}, j = 1, 2, \ldots, n_{\mathrm{O}}$ possible ones, which is time-consuming for a larger base classifier pool. In addition, because some of combinations may have the similar, even same recognition rate, building external ensemble model with the most compact structure is essentially a multi-modal optimization problem. To solve this problem, multi-modal evolutionary algorithm is employed to seek the most satisfactory combination for recognition.

III. Hybrid Base Classifier of Internal Ensemble Learning Model

The main idea of internal ensemble learning model is to seek a sub-classifier with the optimal recognition performance as base classifier. For an imbalanced dataset, a part of majority samples are extracted by traditional under-sampling techniques, and then integrated with data in minority class to form a balanced one. Following that, a sub-dataset, represented by D_i, is derived from the training data available by random selection, and employed to build three sub-classifiers, denoted as $SC_i = \{sc_i^1, sc_i^2, sc_i^3\}$. ELM-, NBM- or KNN-based sub-classifier that has the most satisfactory performance is treated as base classifier for D_i, as shown in Fig. 2.

No matter which kind of sub-classifier is built, grid search, as an exhaustive search method, is employed to seek the optimal values for the key parameters of each sub-classifier by cross validation [31]. Based on three heterogeneous sub-classifiers with optimal parameters, their outputs are synthesized by weighted sum. Since only the sub-classifier with the highest recognition

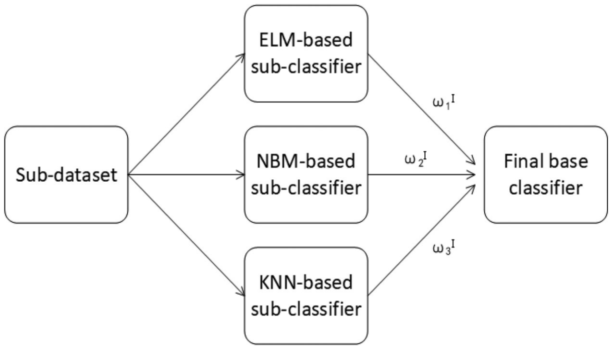


Fig. 2. The structure of internal ensemble learning model.

accuracy is selected as base classifier for a sub-dataset, the weight of ith sub-classifier is defined as $\omega_i^{\mathrm{I}} \in \{0,1\}$, and its value is determined by recognition accuracy. Denoted real and predictive labels of a sub-dataset as y_{I} and $\hat{y}_{\mathrm{I}}$, respectively, the training error of internal ensemble model is defined as F=$\|y_{\mathrm{I}} - \hat{y}_{\mathrm{I}}\|_2$, and its final output is formulated as follows.

$$\hat{y}_{\mathrm{I}} = \sum_{i=1}^{3} \omega_i^{\mathrm{I}} \hat{y}_i, \qquad \text{s.t.} \sum_{i=1}^{3} \omega_i^{\mathrm{I}} = 1 \tag{1}$$

where,

$$\omega_i^{\mathrm{I}} = \begin{cases} 1, i = \arg\max \|\hat{y}_i - y_i\|_2 \\ 0, other \end{cases} \tag{2}$$

A. ELM-Based Sub-Classifier

Extreme learning machine (ELM) employs generalized single hidden layer feed-forward networks(SLFNs) and least-square-based learning algorithm to form a widespread type of feature mappings [31]. Being different from other machine learning algorithms, the input weights and hidden biases of SLFNs are generated randomly and not adjusted during training. Corresponding output weights are analytically determined by generalized inverse operation of output matrix in the hidden layer.

Let $(x_i^j, y_i^j) \in D_i, j = 1, 2, \ldots, |D_i|$ be a sample in ith sub-dataset. Denoted α_k and β_k as the input and output weights, as well as b_k as the biases, a SLFN is formulated as follows.

$$\sum_{k=1}^{L} = \beta_k g(\alpha_k, b_k, x_i^j) = \hat{y}_j \tag{3}$$

In above formula, L represents the number of hidden nodes, and $g(\bullet)$ is the activation function. Let $\beta = [\beta_1^T, \beta_2^T, \ldots, \beta_L^T]^T$, Eq.(3) can be simplified as $H\beta = Y$, where $H = [g(\alpha_k, b_k, x_i^j)]_{|D_i| \times L}$ is output matrix of hidden layer. For a SLFN, α_k and b_k of each hidden node do not change after being randomly initialized. Only output weights are adjusted in terms of the error between the estimated and actual outputs. Denoted δ as a parameter that provides a tradeoff between minimizing the training errors and max-imizing the marginal distance, the output weights are estimated as follows.

$$\hat{\beta} = \begin{cases} (\frac{1}{\delta} + H^T H)^{-1} H^T Y, & l < |D_i| \\ H^T (\frac{1}{\delta} + H^T H)^{-1} Y, & l >= |D_i| \end{cases} \tag{4}$$

B. NBM-Based Sub-Classifier

A Bayesian classifier can predict the probability of a given datum belonging to a specific class. Naive Bayesian Model, as a commonly-used Bayesian classification, is presented based on independent hypothesis of features. That is, the probability of a feature's value in a given class is independent of the values of other features. Based on a sub-dataset, a Naive Bayesian model is built by learning the joint probability distribution of all samples, represented by $P(X_i, Y_i)$. Following that, a datum that needs to be classified will belong to a class that has the largest posterior probability [32].

Let $D_i^k \in D_i$ be a sample of ith sub-datset that belongs to kth class, the prior probability of kth class is estimated as follows.

$$P_k = \frac{D_i^k}{D_i} \tag{5}$$

Because all features on human activity recognition are continuous real number, the conditional probability of a sample follows $p(x_i^j | y_i^k) \sim N(\mu_{kj}, \sigma_{kj}^2)$, where μ_{kj} and σ_{kj} are the mean and variance of kth class. Without loss of generality, it fits for the following probability density function.

$$p(x_i^j | y_i^k) = \frac{1}{\sqrt{2\pi}\sigma_{kj}^2} exp(-\frac{(x_i^j - \mu_{kj})^2}{2\sigma_{kj}^2} \tag{6}$$

C. KNN-Based Sub-Classifier

In a K-nearest neighbor sub-classifier, a sample is labelled as the specific class that most of its neighbors belong to. Denoted x_i^j and x_i^k as inputs of two samples in ith dataset, the distance between them is defined as follows [33].

$$L_p(x_i^j, x_i^k) = (\sum |x_i^j - x_i^k|^p)^{\frac{1}{p}} \tag{7}$$

In the formula, p is a parameter. L_1 means Manhattan distance, whereas L_2 corresponds to the commonly-used Euclidean distance. As $p \longrightarrow \infty$, it represents Chebyshev distance. In this paper, $p = 2$.

IV. Multimodel Evolutionary Optimization for Structure of External Ensemble Learning

In a traditional ensemble classifier, the outputs of all base classifiers are aggregated by weighted sum, with the purposed of improving classification accuracy. Though the weights are dynamically adjusted in terms of the performance of each base classifier, the number of base classifiers is generally set in advance and remains unchanged during training. However, too many base classifiers in an ensemble learning model increases the computation cost. Moreover, the combinations composed of various base classifiers or different number of ones may have the similar, even same classification performance. Thus, seeking the appropriate base classifiers to construct an ensemble learning model is a multimodal optimization problem in essence [34].

To address the challenging issue, multimodal evolutionary algorithms is introduced to dynamically select base classifiers from the ones generated by internal ensemble learning model, with the purpose of forming the most appropriate and compact combination for a specific imbalanced dataset.

Evolutionary algorithms (EAs), as a class of meta-heuristic optimization methods, have the significant advantages in solving non-convex, discontinuous optimization problems [35], thus, have been widely applied to seek the optimal values of key parameters in a classifier [36], [37]. Cheng [38] proposed an adaptive class-specific cost regulation extreme learning machine (CCR-ELM) for class imbalance learning, in which variable-length brain storm optimization algorithm was employed to seek the most rational parameters of CCR-ELM. Luo [39] constructed an optimal sparse spectral clustering based on multi-objective evolutionary algorithm. Godoy [40] optimized radial-based function network by co-evolutionary optimization method. In order to reduce computation cost, Lim [41] introduced EAs to optimize the parameters of a cluster-based ensemble resampling method. Yang [42] selected the most useful samples for training by optimization technique. Rich studies have been done on applying EAs to build an optimal imbalance classifier [43], [44]. One is to extract the most useful data for resampling strategies. Each locus corresponds to a datum. The value of a gene is set to 1 if this datum is assigned for training. However, the length of an individual depends on the size of a dataset, which is infeasible for a large-scale imbalanced dataset. Moreover, its binary-encoding individual is sparse, which slows down convergence. Being different from previous studies, multimodal evolutionary algorithm is introduced to seek the smallest number of base classifiers with highest classification accuracy, with the purpose of building the most compact and optimal structure for external ensemble learning model. Its pseudo-code is provided in Algorithm 1.

As shown in Algorithm 1, an individual is generated by randomly setting some of genes to 1, and its fitness value is evaluated by K-fold cross-validation accuracy. Crossover and mutation operations are done on producing new offspring from parents, and niche strategy is employed to preserve all optima. After integrating all offspring with the current population, the individuals with the best fitness values will be preserved to the population in next generation with a higher probability by roulette wheel selection. All above-mentioned evolutionary operations are repeated until the maximum number of iterations is reached. Apparently, encoding method, objective function, genetic operators and niche strategy are the key issues.

A. Encoding and Objective Function

Encoding is an important step in EAs, through which important variables can be appropriately represented. In this paper, each gene of an individual corresponds to a base classifier and is encoded by binary number, namely 0 or 1. As the value of a specific locus is 1, the corresponding base classifier becomes active and is selected to build an ensemble learning model. Otherwise, this base classifier keeps inactive. Apparently, the length of an individual depends on the number of base classifiers.

Algorithm 1: Seeking the Optimal Structure Based on Multimodal Evolutionary Algorithm.

Require: C_i**:** Base classifiers, N_p**:** Population size, r**:** the niche radius, Len**:** the length of an individual, T**:** Terminal iteration, p_m**:** Mutation probability, p_c**:** Crossover probability, N_A**:** The size of external archive

Input: pop
  Pop $\leftarrow$ Initialization(N_p, Len)
for $i = 1$ to T **do**
    $Pool \leftarrow$ Selection(pop)
    $Pool \leftarrow$ Crossover(pool,p_c)
    $Pool \leftarrow$ Mutation(pool,p_m)
    $Pool \leftarrow$ Niche(pool,r)
    $Pop_n \leftarrow$ [pop;pool]
    $fitness \leftarrow$ Evaluate(pop_n)
    $roulette \leftarrow$ Elite(pop_n,fitness)
    if $len(Archive) >= N_A$ **then**
      $Archive \leftarrow$ dropout($Archive$)
    end if
    $pop =$ choice(pop_n,fitness)
end for

Being different from traditional ensemble learning methods, only part of base classifiers are employed to produce a final label of a sample. Different individuals refers to various combinations of base classifiers.

In external ensemble learning model, the smallest number of base classifiers is activated to form a combination in terms of its classification performance. Thus, G-means of K-fold cross-validation is designed as the fitness function. For an individual, denoted as $\boldsymbol{xc_i}, i = 1, 2, \ldots, N_p$, each gene, represented by $xc_{ik} \in \{0, 1\}, k = 1, 2, \ldots, Len$, corresponds to a base classifier. Thus, the length of an individual equals to the number of base classifiers generated from internal ensemble learning model, namely, $N_B = Len$. For an imbalanced datum, denoted y and $\hat{y}$ as its real label and output of external ensemble learning model, respectively. The training error of ED-ensemble method for this datum is defined as follows.

$$Er = ||\hat{y} - y||_2 \tag{8}$$

B. Genetic Operators and Niche Strategy

During the evolution, offspring are produced based on parents by crossover and mutation operations, respectively. We employ single-point crossover and mutation in the paper due to binary-encoding of an individual. In crossover operation, let $\boldsymbol{xc_i}$ and $\boldsymbol{xc_j}$ be two parents that randomly selected from a population, and p_c be the crossover probability, two offspring, represented by $\boldsymbol{\bar{xc}_i}$ and $\boldsymbol{\bar{xc}_j}$, are formed by swapping the specific genes of two parents in terms of p_c. The detailed crossover process is shown in Eq.(9), where k represents kth loci of two individuals, and a_c is a crossover point randomly generated.

$$\bar{xc}_{ik} = \begin{cases} xc_{ik}, k <= a_c \\ xc_{jk}, k > a_c \end{cases} \tag{9}$$

In mutation operation, the value of a gene randomly selected from $\boldsymbol{xc_i}$ is replaced by the opposite binary number, namely, $\bar{\bar{x}}c_{ik} = 1$ if $xc_{ik} = 0$. Subsequently, a new individual is formed. Based on the above-mentioned genetic operations, the number of base classifiers contained in an ensemble learning model may become larger as more genes change their values from 0 to 1, and vice versa. In some cases, the number of active base classifiers before and after genetic operations may remain the same. However, various base classifiers are chosen to construct a combination, resulting in different recognition performances.

To the best of our knowledge, there are more than one local and global optima in a multimodal optimization problem. In order to avoid falling into the local optima, a niche technique is introduced to maintain the diversity of a population [45]. During the evolution, each optimum refers to a niche, and all peaks are stably covered by sub-populations, with the purpose of providing the selection pressure within the local search space [46]. Rich studies have been done on various niching methods, such as clearing, clustering, crowding and sharing [47]. We employ sharing strategy to find each niche and form a sub-population in the paper. Let r be the niche radius, and $d_{ij} = ||\boldsymbol{xc_i} - \boldsymbol{xc_j}||_2$ be the Euclidean distance between two individuals, two individuals are similar if $d_{ij} <= r$ is satisfied. Once $d_{ij} > r$, the individual with the worst fitness value is penalized by reducing its fitness value.

C. Decision-Making of an Optimal Structure

Without loss of generality, more than one combination is obtained from the above-mentioned multimodal evolutionary algorithm. Among them, the one containing the larger number of base classifiers consumes more computation time and resources for training. Thus, we prefer the optimal combination with the most compact structure. Denoted N_B as the number of base classifiers, and P_{best} as the optimal solutions, an individual $\boldsymbol{xc_i} \in P_{best}$ that contains the smallest number of '1' is defined as the most satisfactory optimum, namely, $i = \arg\min_j \sum_{j=1}^{|P_{best}|} Le(\boldsymbol{xc_j})$. Here, $Le(\boldsymbol{xc_j}) = \sum_{k=1}^{Len} xc_{jk}$ represents the total number of active base classifiers.

Let ω_i^{O} be the weight of ith base classifier, the final output of an ensemble classification method is the weighted sum of all active base classifiers. It is worth noting that $\omega_i^{\mathrm{O}} = 0$ for inactive base classifiers.

$$\hat{y} = \sum_{i=1}^{N_B} \omega_i^{\mathrm{O}} \hat{y}_i \quad \text{s.t.} \sum_{i=1}^{N_B} \omega_i^{\mathrm{O}} = 1 \tag{10}$$

where

$$\omega_i^{\mathrm{O}} = \frac{1}{N_B} \tag{11}$$

V. Experimental Study

To verify the effectiveness of the proposed ED-ensemble method for human activity recognition, 7 imbalanced datasets derived from Localization Data for Person Activity (LDPA) are employed as the benchmarks. All of them are binary classification problems, and the detailed specifications, such as imbalance ratio (IR), are listed in Table I. LDPA built by Mitja Lustrek in 2010 records the monitoring data of real-world human activities that are collected from four sensors laid on left ankle, right ankle, belt and bust of a person [48]. There are 8 attributes and 11 human behaviors are recorded, including walking, falling, lying down, lying, sitting down, sitting, standing up from lying, on all fours, sitting on the ground, standing up from sitting, standing up from sitting on the ground. All experiments have implemented on AMD ryzen 7 2700x eight core processor 3.70 GHz, 16 GB ram and NVIDIA Titan XP with Python 3.5.

Three groups of experiments have been done to observe the rationality of the proposed framework. Base classifiers built by different learning models are compared in Experiment I. In Experiment II, the multimodal characteristic of ED-ensemble model is proved, and the feasibility of decreasing the number of base classifiers is demonstrated. Experiment III is to compare the proposed ED-ensemble with the other class imbalance learning methods.

In ED-ensemble method, ELM, NBM and KNN are employed to build sub-classifiers and their main parameters are set as follows: the number of hidden neurons in ELM is 25 and sigmoidal function is used as the activation function; the Laplacian smoothing coefficient and the prior probability coefficient in NBM are set to 0 and 1, respectively; the range of k and p in KNN are [1,11] and $[1, \infty)$. Moreover, the main parameters of multi-modal evolutionary algorithm are set as follows: the length of each individual $Len = 30$, population size $N_p = 50$, the crossover probability $p_c = 0.6$, the mutation probability $p_m = 0.08$ and the maximum number of iterations $T = 100$.

A. The Evaluation Metrics

Five metrics are employed to provide more comprehensive evaluation for recognition accuracy. Denoted TP, TN, FP and FN as the number of data with true positive, true negative, false positive and false negative, respectively, Accuracy, as the most commonly-used metric, is defined as follows.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{12}$$

However, Accuracy cannot completely describe the classification accuracy for the minority data that are misclassified to the majority class. Thus, G-mean and the area under the curve (AUC) are introduced. AUC is calculated by the area under the receiver operating characteristic (ROC) graph. Based on Recall that refers to the number of minority data being correctly classified, represented by $TP/(TP + FN)$, G-mean is defined as follows.

$$G - mean = \sqrt{\frac{TP}{TP + FN} \times \frac{TN}{TN + FP}} \tag{13}$$

Denoted $TP/(TP + FP)$ as Precision, F-value calculates the harmonic mean of Precision and Recall, with the purpose of

TABLE I
BINARY-CLASS IMBALANCED BENCHMARK DATASETS FROM LDPA

Datasets	IR	Training Data	Test Data	Actions
FLD	2.2	383	257	Falling,lying down
LDSD	3.7	333	223	Lying down,sitting down
LOAF	10.4	2615	1742	Lying,on all fours
LS	2	3567	2377	Lying,sitting
SUFLSUFS	14.7	847	564	Standing up from lying,standing up from sitting
SUFSSUFSOG	2.4	183	122	Standing up from sitting,Standing up from sitting on the ground
FSD	9.25	193	129	Falling,sitting down

emphasizing the performance on the minority samples.

$$F-value = \frac{\left(1+\beta_F^2\right)\cdot Recall \cdot Precision}{\beta^2(Recall + Precision)} \quad (14)$$

For an ensemble classification method, average Q-statistic comprehensively reflects the diversity of all base classifiers. A larger value of Q_{av} means a worse diversity among them [49]. Assuming that M^{11} and M^{00} represent the number of data being correctly classified and misclassified by both classifiers, M^{01} and M^{10} are their number of data being misclassified. Q-statistic is defined to compare the difference between two base classifiers [28].

$$Q_{\alpha v} = \frac{2}{N(N-1)}\sum_{i=1}^{N-1}\sum_{k=i+1}^{N} Q_{ik} \quad (15)$$

$$Q_{ik} = \frac{M^{11}M^{00} - M^{01}M^{10}}{M^{11}M^{00} + M^{01}M^{10}} \quad (16)$$

B. Comparison of Recognition Accuracy on Various Base Classifiers

To the best of our knowledge, sub-classifiers with various structures sketch the distribution of corresponding sub-dataset more comprehensively. To further verify the necessity of a hybrid base classifier, the proposed internal ensemble learning model is compared with the ones composed of homogeneous sub-classifiers. ELM, NBM and KNN are employed to build sub-classifiers, and the number of sub-classifiers is preset to 3. Based on the original imbalanced dataset, five groups of sub-datasets, denoted as RUS1, RUS2, ..., RUS5, are generated by under-sampling with different sampling rates, including 10%, 25%, 50%, 75% and 90%, respectively. For these sub-datasets, the diversity of ELM-, NBM-, KNN-based homogeneous ensemble learning models are compared with that of the proposed heterogeneous one by the values of Q-statistic, as listed in Table II. Apparently, a hybrid base classifier provides the smallest values of Q-statistics for all sub-datasets, indicating the most distinct difference among sub-classifiers. Among homogeneous base classifiers, we observe that ELM-based ensemble learning model is inferior to the others in diversity due to the largest Q-statistic values.

TABLE II
COMPARISON OF Q-STATISTIC AMONG VARIOUS BASE CLASSIFIERS

Under-sampling	ELM	NBM	KNN	Hybrid
RUS1	0.539	0.518	0.527	**0.486**
RUS2	0.528	0.509	0.517	**0.492**
RUS3	0.534	0.517	0.533	**0.464**
RUS4	0.531	0.526	0.526	**0.477**
RUS5	0.527	0.505	0.545	**0.452**

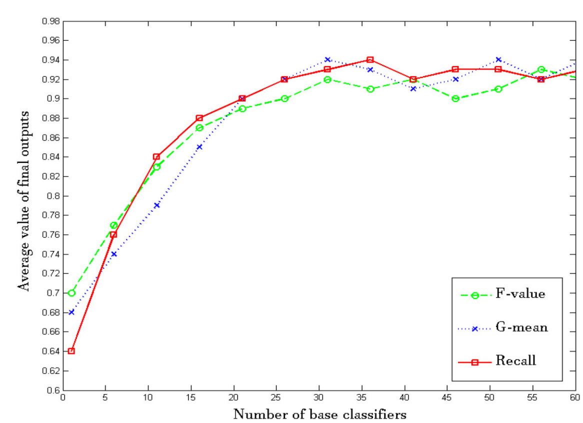


Fig. 3. The classification performances of ED-ensemble algorithm under different N_B.

In an ensemble imbalance classifier, the number of base classifiers plays the direct impact on recognition accuracy. Thus, we further analyze recognition performances of human actions under different N_B. By comparing average G-mean, F-values and Recall of the proposed ED-ensemble method as N_B varies from 1 to 50, we see from Fig. 3 that with the increasing of N_B, the proposed dual-ensemble learning model can recognize various human actions with accuracy higher and higher. More especially, all metrics are stable up to 90% as the number of base classifiers exceeds 30. The reason for this tendency is that too few base classifiers cannot fully sketch the distribution of imbalanced datasets, resulting in misclassification, especially for

TABLE III
THE OPTIMAL COMBINATIONS OF ED-ENSEMBLE MODEL

No. of optima	RUS1		RUS2		RUS3		RUS4		RUS5	
	AUC	Number	AUC	Number	AUC	Number	AUC	Number	AUC	Number
1#	0.951	20	0.9526	17	0.9623	18	**0.9692**	**17**	0.9737	20
2#	0.9505	19	0.9526	17	0.9612	16	0.9726	19	**0.9737**	**15**
3#	0.9505	17	**0.9526**	**15**	**0.9626**	**16**	0.9692	18	0.9737	19
4#	**0.9512**	**19**	0.9507	18	0.9623	20	0.9692	18	0.9737	17

TABLE IV
BASE CLASSIFIERS FOR EXTERNAL ENSEMBLE MODEL

No.	RUS1		RUS2		RUS 3		RUS4		RUS5	
	AUC	Classifier	AUC	Classifier	AUC	Classifier	AUC	Classifier	AUC	Classifier
1# base classifier	**0.8953**	**ELM**	0.8522	ELM	**0.8845**	**NBM**	**0.9354**	**KNN**	0.9216	ELM
2# base classifier	**0.9767**	**ELM**	**0.9151**	**NBM**	**0.9245**	**KNN**	**0.9135**	**ELM**	**0.9471**	**ELM**
3# base classifier	**0.9534**	**KNN**	0.7212	KNN	0.8461	KNN	**0.9468**	**ELM**	**0.9835**	**NBM**
4# base classifier	**0.9165**	**NBM**	0.8326	NBM	**0.9135**	**NBM**	**0.9325**	**KNN**	0.8522	ELM
5# base classifier	**0.8604**	**ELM**	**0.9333**	**NBM**	0.8148	ELM	0.8466	NBM	**0.9742**	**ELM**
6# base classifier	**0.9722**	**ELM**	0.8544	ELM	**0.9146**	**ELM**	0.8654	NBM	**0.9487**	**KNN**
7# base classifier	**0.8953**	**NBM**	**0.9166**	**KNN**	0.7135	NBM	**0.9351**	**NBM**	0.8846	ELM
8# base classifier	**0.9069**	**ELM**	**0.9633**	**KNN**	0.7659	ELM	0.7684	ELM	0.8412	ELM
9# base classifier	**0.9534**	**NBM**	**0.9454**	**ELM**	0.8156	KNN	0.8216	ELM	**0.9882**	**KNN**
10# base classifier	0.7786	KNN	**0.9317**	**NBM**	**0.9645**	**KNN**	**0.9354**	**NBM**	**0.9785**	**ELM**
11# base classifier	**0.9534**	**KNN**	**0.9437**	**KNN**	**0.9133**	**KNN**	0.8978	KNN	0.8456	ELM
12# base classifier	**0.9006**	**ELM**	**0.9190**	**NBM**	**0.9483**	**ELM**	0.7988	KNN	0.8412	ELM
13# base classifier	**0.9354**	**NBM**	**0.8924**	**NBM**	0.8165	ELM	0.7512	ELM	**0.9854**	**NBM**
14# base classifier	0.8522	ELM	**0.9378**	**NBM**	**0.8648**	**NBM**	0.8214	NBM	0.7564	KNN
15# base classifier	0.7625	ELM	0.8281	NBM	**0.9354**	**ELM**	**0.9342**	**NBM**	**0.9512**	**NBM**
16# base classifier	**0.9398**	**NBM**	0.9318	ELM	0.8146	ELM	**0.9336**	**NBM**	**0.9621**	**ELM**
17# base classifier	0.8225	ELM	0.7945	ELM	**0.8646**	**NBM**	**0.9787**	**ELM**	0.8787	ELM
18# base classifier	**0.9067**	**KNN**	0.8641	KNN	0.7125	ELM	0.8687	NBM	**0.9587**	**NBM**
19# base classifier	**0.8836**	**ELM**	0.8799	ELM	**0.9138**	**ELM**	0.8684	NBM	0.939	KNN
20# base classifier	0.7509	NBM	**0.9718**	**ELM**	0.8154	ELM	**0.9642**	**ELM**	**0.9512**	**ELM**
21# base classifier	**0.9134**	**ELM**	**0.9553**	**KNN**	0.7348	KNN	0.7668	KNN	0.9145	NBM
22# base classifier	0.8514	KNN	**0.8984**	**NBM**	**0.9463**	**ELM**	**0.9785**	KNN	0.9116	NBM
23# base classifier	0.8677	ELM	0.8501	ELM	**0.8646**	**ELM**	0.8436	ELM	**0.9792**	**NBM**
24# base classifier	**0.9154**	**ELM**	0.8666	KNN	0.8484	NBM	**0.9484**	**ELM**	**0.9587**	**ELM**
25# base classifier	**0.9354**	**KNN**	0.8401	KNN	**0.9354**	**KNN**	**0.9786**	**NBM**	0.9472	ELM
26# base classifier	**0.8625**	**KNN**	0.8295	NBM	0.7848	ELM	0.8135	KNN	**0.9621**	**KNN**
27# base classifier	0.7522	NBM	0.8659	ELM	0.8135	KNN	0.7656	KNN	0.8901	ELM
28# base classifier	0.7937	NBM	**0.9654**	**KNN**	**0.9155**	**ELM**	**0.9846**	**KNN**	0.9268	NBM
29# base classifier	0.8604	ELM	0.8143	ELM	**0.9460**	**NBM**	**0.8457**	**ELM**	**0.9636**	**KNN**
30# base classifier	0.8509	NBM	**0.9151**	**NBM**	0.8134	ELM	**0.9534**	**KNN**	0.9024	KNN

TABLE V
COMPARISON OF THE CLASSIFICATION PERFORMANCE AMONG VARIOUS CLASSIFICATION METHODS

Method	Accuracy	G-mean	F1-score	Recall	AUC
SVM	0.9801±0.0651	0.8835±0.1154	0.8707±0.0248	0.7911±0.2089	0.8976±0.1024
DT	0.9727±0.0248	0.8771±0.0122	0.8276±0.1704	0.8218±0.0053	0.8868±0.1134
MLP	0.9783±0.0162	0.9028±0.0384	0.8496±0.1085	**1.0±0.0**	0.8960±0.0798
ELM	0.9737±0.0263	0.8535±0.0465	0.8069±0.1614	0.8734±0.0210	0.8021±0.1667
NBM	0.9413±0.0717	0.7667±0.2014	0.8435±0.1565	0.8952±0.0802	0.8565±0.1445
KNN	0.9621±0.0379	0.7727±0.2273	0.7714±0.1232	0.6544±0.2486	0.8272±0.1728
Bagging	0.9828±0.0138	0.9165±0.0137	0.8677±0.1361	**1.0±0.0**	0.8672±0.1158
UnderBagging	0.9763±0.0172	0.9262±0.0074	0.9210±0.0709	0.8583±0.1417	0.9289±0.0711
ANASYN-Bagging	0.9644±0.0025	0.9355±0.0199	0.8593±0.0785	0.7208±0.1542	0.8593±0.0724
AdaBoost	0.9611±0.0951	0.9203±0.0071	0.8697±0.0725	0.7292±0.4458	0.8531±0.2259
Random Forest	0.9768±0.0181	0.9510±0.0444	0.8798±0.0536	0.8091±0.0084	0.9032±0.0344
EE-Bagging	0.9808±0.0079	0.9475±0.0236	0.8954±0.0602	0.8044±0.0489	0.9093±0.0249
ED-ensemble	**0.9872±0.0012**	**0.9862±0.0033**	**0.9766±0.0085**	**1.0±0.0**	**0.9876±0.0064**

the minority data. Based on this, the number of base classifiers is set to 30 in the following experiments.

C. Analysis of Multimodel Characteristics of External Ensemble Learning Model

Under RUS resampling strategies with different sampling rates as 10%, 25%, 50%, 75%, 90%, a particular number of base classifiers are selected to participate in aggregation. In order to compare the overall recognition accuracy, the following experiments are designed. Under each under-sampling strategy, 30 base classifiers are trained separately. Subsequently, 10, 20, and 30 base classifiers are randomly selected to form ensemble models, respectively. Fig. 4 depicts their overall accuracy. Apparently, ensemble learning models composed of the same number of base classifiers may have different overall accuracy because base classifiers have various parameters, and there may be more than one ensemble learning model obtaining the optimal recognition accuracy. Under the different number of base classifiers, recognition accuracy may not become better, sometimes be similar, with the increasing of base classifiers. Thus, classification performance of an ensemble learning method is more sensitive to its construction and it is necessary to find the combination of base classifiers with the most compact structure and best recognition accuracy.

In ED-ensemble method, a part of base classifiers are extracted to construct the optimal combination of external ensemble learning model. We observe from Table III that the combinations with different number of base classifiers may have the same recognition performances. Thus, ensemble class imbalance learning model is in essence multimodal. To be specific, four kinds of combinations for RUS5 all recognize human actions with the same AUC as 0.9737, but consist of different number of base classifiers. Apparently, the first combination contains more redundant base classifiers than the others.

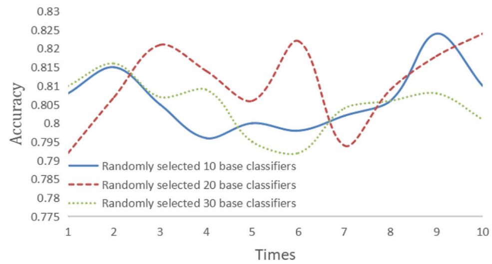
Table IV lists all base classifiers derived from internal ensemble learning model and their AUC values under different under-sampling rates. Among them, the bold ones compose of the optimal ED-ensemble model with the smallest number of base classifiers and the highest recognition accuracy after evolution.

D. Comparion of Recognition Performance Among Different Classification Methods

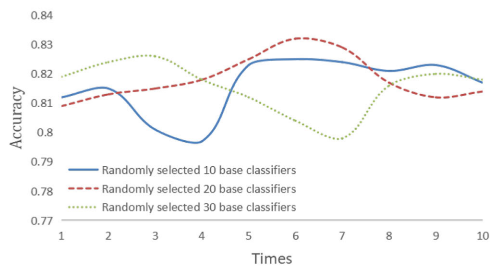
Seven widely-used classification methods, including Bagging, SVM, DT, MLP, NBM, ELM, KNN, and five ensemble class imbalance learning methods, such as UnderBagging, ANASYN-Bagging, AdaBoost, Random Forest and Easy Ensemble Bagging (EE-bagging), are employed as the comparative ones. Fig. 5 depicts recognition performances of seven traditional classification methods with single classifier for imbalanced datasets from LDPA. Apparently, ELM-based classifier is inferior to the others in all metrics. It is worth noting that Accuracy for all of methods are similar and higher than 95%, which is not in accordance with the tendency of other metrics. This phenomenon indicates that Accuracy cannot exactly evaluate the classification performance of class imbalance learning methods.

As far as we know, ensemble class imbalance learning methods synthesize the output of more than one base classifier to provide a global sight for the skewed dataset. Fig. 6 and Fig. 7 depict recognition performances of the proposed ED-ensemble and the other five ensemble classification methods for LDPA dataset. Also, recognition performances of seven single classifiers and six ensemble classification methods for human activities are compared in Table V. Apparently, ED-ensemble method outperforms the others on all metrics significantly due to various base classifiers obtained from internal ensemble model and the optimal structure found by multimodal evolutionary algorithm.

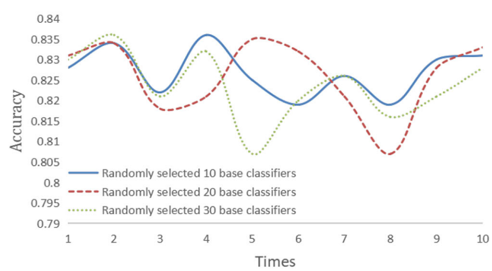
The statistical experimental results listed in Table V show that UnderBagging is superior to the other ensemble class imbalance learning methods, except for ED-ensemble proposed in the paper. In order to further compare the recognition accuracy of



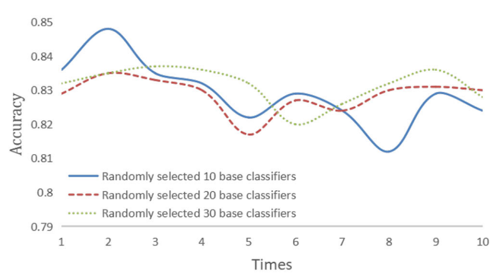
(a)



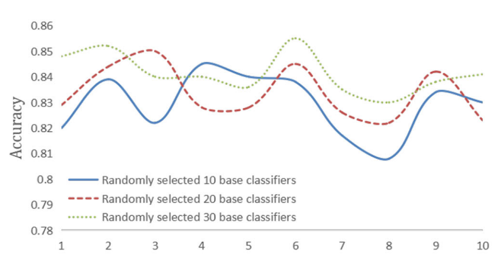
(b)



(c)



(d)



(e)

Fig. 4. Multi-modal characteristics under different RUS (sampling rate=10%, 25%, 50%, 75%, 90%).

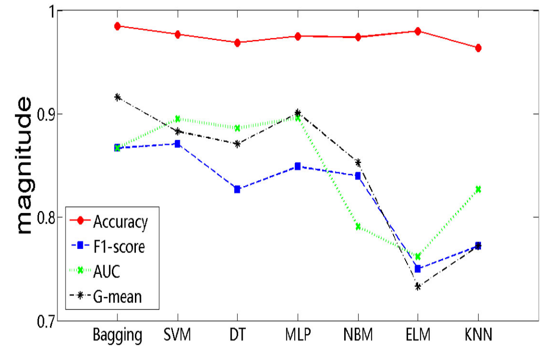


Fig. 5. Comparison of recognition performances among different traditional classification methods.

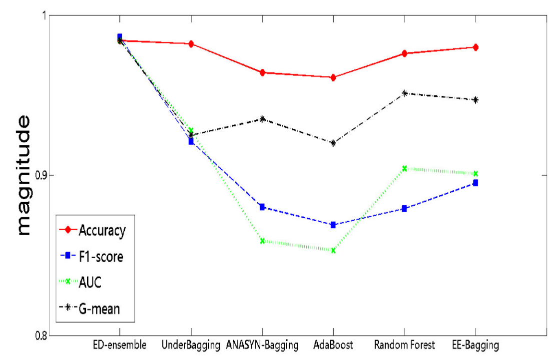


Fig. 6. Comparison of the classification performance among different ensemble class imbalance learning methods.

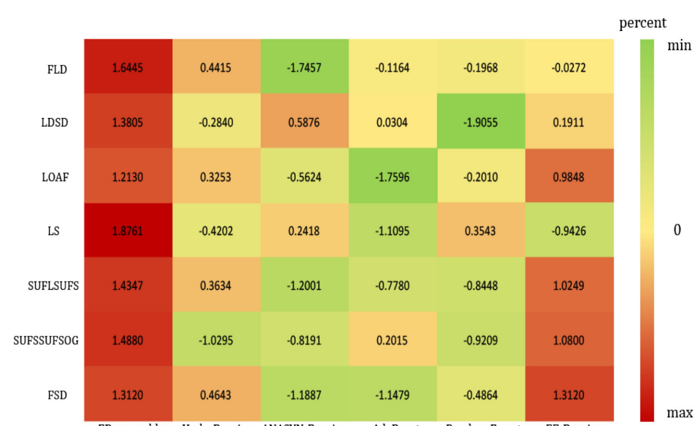


Fig. 7. Comparison of G-mean among different ensemble class imbalance learning methods under 7 LDPA datasets.

UnderBagging and ED-ensemble methods, G-means obtained by them under various under-sampling rates are listed in Table VI and Table VII. We see that no matter how many majority samples are selected from original imbalanced dataset, ED-ensemble identifies human actions more exactly because base classifier derived from more than one heterogeneous sub-classifier can grasp the specific distribution of each sub-dataset and the rational combinations effectively guarantee the recognition accuracy.

TABLE VI
COMPARISON OF AUC AMONG DIFFERENT ENSEMBLE CLASSIFICATION METHODS

Datasets	ED-ensemble	UnderBagging	ANASYN-Bagging	AdaBoost	Random Forest	EE-Bagging
FLD	**0.9701±0.0012**	0.9162±0.0544	0.8182±0.0785	0.8912±0.0203	0.8876±0.0124	0.8952±0.0802
LDSD	**0.9988±0.0012**	0.9522±0.0396	0.9766±0.0264	0.9610±0.0312	0.9068±0.0306	0.9655±0.0299
LOAF	**0.9521±0.0151**	0.9128±0.0183	0.8735±0.0122	0.8205±0.0504	0.8895±0.0308	0.9420±0.0317
LS	**0.9772±0.0158**	0.8506±0.1533	0.8871±0.1103	0.8126±0.0638	0.8933±0.0267	0.8218±0.0053
SUFLSUFS	**0.9533±0.0096**	0.8571±0.0454	0.7167±0.1365	0.7546±0.0154	0.7486±0.0459	0.9165±0.0141
SUFSSUFSOG	**0.9211±0.0190**	0.6268±0.1406	0.6514±0.1255	0.7707±0.0248	0.6395±0.1507	0.8734±0.0210
FSD	**0.9808±0.0381**	0.8811±0.0813	0.6867±0.1365	0.6915±0.0184	0.7693±0.1359	0.9496±0.0651
Average Rank	**1.2500**	3.9750	4.1500	4.7000	5.3750	3.6500
Win-lose-draw	-	5-1-1	4-0-3	4-1-2	4-2-1	5-0-2
Holm-Post-Hoc Test	-	**2.54631E-07**	**2.84613E-07**	**1.65492E-07**	**5.78946E-08**	**3.16134E-07**

TABLE VII
G-MEAN OF ED-ENSEMBLE WITH VARIOUS UNDER-SAMPLING RATES

Datasets	RUS1	RUS2	RUS3	RUS4	RUS5	Mean
FLD	0.9007±0.0225	0.9558±0.0112	0.9506±0.0121	0.9541±0.0270	**0.9664±0.0232**	0.9721±0.0210
LDSD	**0.9835±0.0157**	0.9450±0.0129	0.9715±0.0076	0.9555±0.0016	0.9753±0.0120	0.9772±0.0167
LOAF	0.8311±0.0461	0.8624±0.0711	0.8721±0.0797	0.9363±0.0122	**0.9885±0.0225**	0.9468±0.0036
LS	0.9622±0.0222	0.9693±0.0322	0.9260±0.0014	**0.9855±0.0016**	0.9340±0.0033	0.9760±0.0226
SUFLSUFS	0.8725±0.0234	0.9537±0.0304	0.8754±0.0142	**0.9767±0.0129**	0.9161±0.0172	0.9477±0.0105
SUFSSUFSOG	0.8006±0.0299	0.8598±0.1079	0.8622±0.0171	**0.9354±0.0420**	0.8622±0.0178	0.9501±0.0360
FSD	**0.9793±0.0299**	0.9276±0.0184	0.9260±0.0080	0.9696±0.0199	0.9632±0.0166	0.9699±0.0254

TABLE VIII
G-MEAN OF UNDERBAGGING WITH VARIOUS UNDER-SAMPLING RATES

Datasets	RUS1	RUS2	RUS3	RUS4	RUS5	Mean
FLD	0.4332±0.1269	0.4586±0.0205	0.5075±0.0979	**0.5581±0.2473**	0.5085±0.0150	0.4876±0.0925
LDSD	0.6127±0.3960	**0.7835±0.0072**	0.6780±0.0901	0.7571±0.0419	0.6887±0.0644	0.7172±0.0652
LOAF	0.4152±0.2758	0.4694±0.4106	0.4667±0.3391	**0.5877±0.3952**	0.4152±0.2547	0.4688±0.2784
LS	0.6906±0.1657	**0.7669±0.1128**	0.6917±0.1370	0.7568±0.0177	0.6609±0.1657	0.7068±0.1391
SUFLSUFS	0.8144±0.0856	**0.9692±0.0072**	0.9306±0.0191	0.9464±0.0419	0.8144±0.0644	0.9032±0.0397
SUFSSUFSOG	0.6887±0.0384	**0.7749±0.1128**	0.5636±0.0208	0.4740±0.2560	0.4710±0.2171	0.5895±0.1391
FSD	0.8090±0.1555	0.8229±0.0471	0.8588±0.0481	0.8401±0.0467	**0.8632±0.0394**	0.8408±0.0412

VI. CONCLUSION

Human activity recognition is an important and hot topic in many real-world applications. Since various human actions may occur at different frequencies, the corresponding monitoring data is skewed. To solve this imbalance classification problems, a dual-ensemble class imbalance learning method is presented, in which two ensemble learning models are nested each other. Three heterogeneous sub-classifiers compose of internal ensemble learning model, and the one with the highest recognition accuracy is selected as base classifier for the corresponding sub-dataset. In external ensemble learning model, an optimal combination that contains the smallest number of base classifiers and identifies human actions most exactly is found by multimodal evolutionary algorithm. Experimental results on seven imbalanced activity datasets derived from LDPA show that the proposed ED-ensemble outperforms the other five ensemble classification methods in all metrics. Applying ED-ensemble method proposed in the paper to other imbalanced scenarios, such as fault diagnosis of industrial facilities will be our future works.

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